Corporate Liquidity and Debt Buybacks

Zhiguo He, Jessica Shi Li, Qiping Xu

Discussion by:

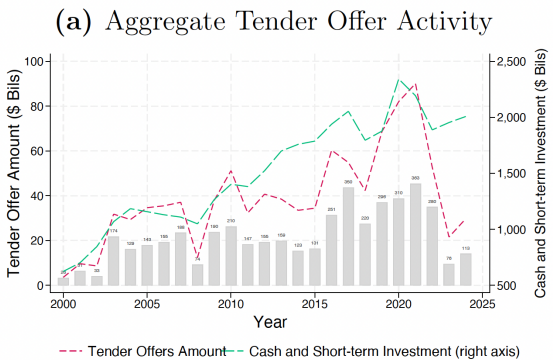
Zhiyang Luo

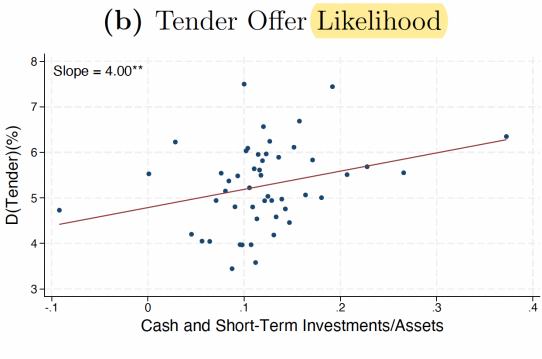
Pre-Seminar 2026

Context: Debt Buybacks

Firms retire their bonds before maturity, and they pay the market price.

- 4,347 verified tender offers, $1.08 trillion retired over 2000–2024





In the data, corporate bond buybacks track corporate cash.

Paper in One Slide

RQ: Why do firms buy back their corporate debt?

The literature suggests that they shouldn't (Admati et al. 2018; DeMarzo and He 2021).

- Leverage ratchet effect: equity holders have an incentive to increase debt but are reluctant to reduce it.

Paper in One Slide

RQ: Why do firms buy back their corporate debt?

The literature suggests that they shouldn't (Admati et al. 2018; DeMarzo and He 2021).

- Leverage ratchet effect: equity holders have an incentive to increase debt but are reluctant to reduce it.

DeMarzo and He (2021) (DH) show that equity holders issue debt continuously and **never buy it back**, no matter how levered the firm is.

Paper in One Slide

RQ: Why do firms buy back their corporate debt?

The literature suggests that they shouldn't (Admati et al. 2018; DeMarzo and He 2021).

- Leverage ratchet effect: equity holders have an incentive to increase debt but are reluctant to reduce it.

DeMarzo and He (2021) (DH) show that equity holders issue debt continuously and **never buy it back**, no matter how levered the firm is.

This paper **generates buybacks** in a DH-type model by adding cash liquidity concern.

- key friction: firms cannot freely borrow $\rightarrow$ liquidation is costly

Paper in One Slide

RQ: Why do firms buy back their corporate debt?

The literature suggests that they shouldn't (Admati et al. 2018; DeMarzo and He 2021).

- Leverage ratchet effect: equity holders have an incentive to increase debt but are reluctant to reduce it.

DeMarzo and He (2021) (DH) show that equity holders issue debt continuously and **never buy it back**, no matter how levered the firm is.

This paper **generates buybacks** in a DH-type model by adding cash liquidity concern.

- key friction: firms cannot freely borrow $\rightarrow$ liquidation is costly

Empirically, data confirm the model's predictions: cash-rich firms generally buy back more, and firms adopt debt and dividend payout policies as the model predicts.

Static Model

- Two periods ($t = 0, 1$), no discounting

Static Model

- Two periods ($t = 0, 1$), no discounting

- The firm needs to repay debt of F from cash flows $\tilde{W}$ at $t = 1$.
  - if $W \geq F$, the firm repays debt from cash;
  - if $W < F$, it can either continue by liquidating assets or default.

Static Model

- Two periods ($t = 0, 1$), no discounting

- The firm needs to repay debt of F from cash flows $\tilde{W}$ at $t = 1$.
  - if $W \geq F$, the firm repays debt from cash;
  - if $W < F$, it can either continue by liquidating assets or default.

1. Liquidating partially will damage the going-concern value

$$A + \underbrace{\theta(W - F)}_{<0}, \quad \theta > 1.$$

Static Model

- Two periods ($t = 0, 1$), no discounting

- The firm needs to repay debt of F from cash flows $\tilde{W}$ at $t = 1$.
  - if $W \geq F$, the firm repays debt from cash;
  - if $W < F$, it can either continue by liquidating assets or default.

1. Liquidating partially will damage the going-concern value

$$A + \underbrace{\theta(W - F)}_{<0}, \quad \theta > 1.$$

2. Default gives the cash back to equity holders:

$$W$$

Static Model

- Two periods ($t = 0, 1$), no discounting

- The firm needs to repay debt of F from cash flows $\tilde{W}$ at $t = 1$.
  - if $W \geq F$, the firm repays debt from cash;
  - if $W < F$, it can either continue by liquidating assets or default.

1. Liquidating partially will damage the going-concern value

$$A + \underbrace{\theta(W - F)}_{<0}, \quad \theta > 1.$$

2. Default gives the cash back to equity holders:

$$W$$

- Continue iff

$$A + \theta(W - F) \geq W \quad \Rightarrow \quad \boxed{W \geq W_d \equiv \frac{\theta F - A}{\theta - 1}.}$$

Static Model: Equity Payoff

- Date-1 equity payoff:

$$V^1(W;F) = \begin{cases} W, & W < W_d, \\ \theta(W-F)+A, & W_d \le W < F, \\ W-F+A, & W \ge F. \end{cases} \tag{2}$$

Static Model: Equity Payoff

- Date-1 equity payoff:

$$V^1(W;F) = \begin{cases} W, & W < W_d, \\ \theta(W-F)+A, & W_d \le W < F, \\ W-F+A, & W \ge F. \end{cases} \tag{2}$$

- Marginal value of cash:

$$V^1_W(W;F) = \begin{cases} 1, & W < W_d, \\ \theta, & W_d \le W < F, \\ 1, & W \ge F. \end{cases}$$

Static Model: Equity Payoff

- Date-1 equity payoff:

$$V^1(W;F) = \begin{cases} W, & W < W_d, \\ \theta(W-F)+A, & W_d \le W < F, \\ W-F+A, & W \ge F. \end{cases} \tag{2}$$

- Marginal value of cash:

$$V^1_W(W;F) = \begin{cases} 1, & W < W_d, \\ \theta, & W_d \le W < F, \\ 1, & W \ge F. \end{cases}$$

- Cash has a liquidity premium (for **shareholders**) only in the partial-liquidation region:

$$V_W = \theta > 1.$$

Static Model: Debt Pricing and Date-0 Equity Value

- Define $d \equiv \Pr(\tilde{W} < W_d), \ell \equiv \Pr(W_d \leq \tilde{W} < F), s \equiv \Pr(\tilde{W} \geq F)$.

- Risk-neutral lenders price debt while considering the default risk

$$p(F) \equiv \Pr(\text{do not default}) = \Pr(\tilde{W} \geq W_d) = \ell + s < 1.$$

Static Model: Debt Pricing and Date-0 Equity Value

- Define $d \equiv \Pr(\tilde{W} < W_d), \ell \equiv \Pr(W_d \leq \tilde{W} < F), s \equiv \Pr(\tilde{W} \geq F)$.

- Risk-neutral lenders price debt while considering the default risk

$$p(F) \equiv \Pr(\text{do not default}) = \Pr(\tilde{W} \geq W_d) = \ell + s < 1.$$

- Date-0 equity value:

$$V^0(F) = \mathbb{E}[V^1(\tilde{W}; F)].$$

- The marginal value of reducing debt is:

$$\boxed{-V_F^0 = \theta\ell + s.} \tag{5}$$

Intuition:

retiring one unit of face is worth

0 in default states, θ in partial-liquidation states, 1 in survival states.

Static Model: Intuition

Q: What is the marginal value of hoarding an additional dollar ($W \to W + \Delta w$) on the B/S?

$$\left.\frac{\partial V^{0,\text{hoard}}}{\partial w}\right|_{w=0} = d \cdot 1 + \ell \cdot \theta + s \cdot 1 = 1 + (\theta - 1)\ell \geq 1. \tag{7}$$

Static Model: Intuition

Q: What is the marginal value of hoarding an additional dollar ($W \to W + \Delta w$) on the B/S?

$$\left.\frac{\partial V^{0,\text{hoard}}}{\partial w}\right|_{w=0} = d \cdot 1 + \ell \cdot \theta + s \cdot 1 = 1 + (\theta - 1)\ell \geq 1. \tag{7}$$

Q: What is the marginal value of spending a dollar on buybacks?

$$\text{one dollar retires } \frac{1}{p} \text{units of debt} \times (-V_F^0) = \frac{\theta\ell + s}{\ell + s}$$

Static Model: Intuition

Q: What is the marginal value of hoarding an additional dollar ($W \to W + \Delta w$) on the B/S?

$$\left.\frac{\partial V^{0,\text{hoard}}}{\partial w}\right|_{w=0} = d \cdot 1 + \ell \cdot \theta + s \cdot 1 = 1 + (\theta - 1)\ell \geq 1. \tag{7}$$

Q: What is the marginal value of spending a dollar on buybacks?

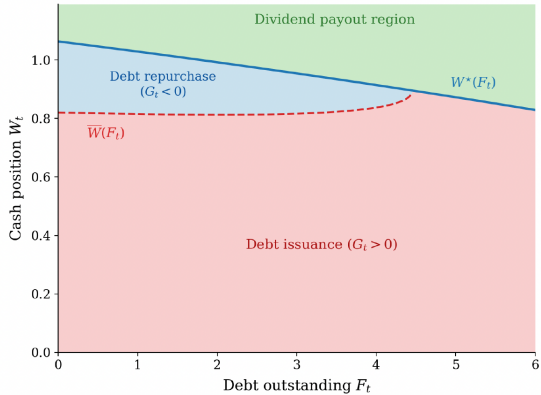
$$\text{one dollar retires } \frac{1}{p} \text{units of debt} \times (-V_F^0) = \frac{\theta\ell + s}{\ell + s}$$

$\Rightarrow$ **Buyback is more valuable** when $\theta > 1$.

$$\frac{\theta\ell + s}{\ell + s} - [1 + (\theta - 1)\ell] = \frac{d}{1 - d}\ell(\theta - 1) > 0.$$

Economically, equity holders value cash more than lenders in the partial liquidity region ($\theta > 1$) $\Rightarrow$ swap is valuable

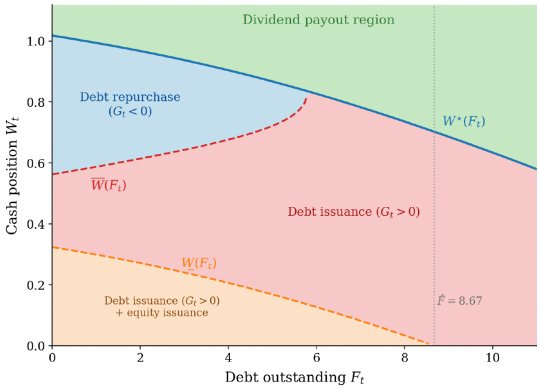
Dynamic Model: Optimal Debt Policy



1. Below the dividend payout boundary, $V_W > 1$: never pay out dividends
   - Cash buffer provides valuable insurance against liquidity-driven bankruptcy.
2. Cash is relatively abundant: buy back debt, since the cash-carry wedge dominates.

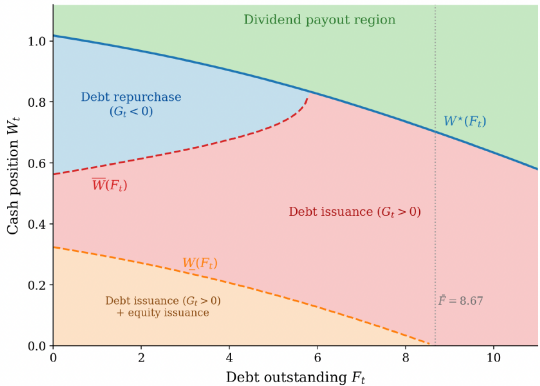
Dynamic Model that Allows for Equity

Main message: debt buyback mechanism **still holds** with external equity financing.



Dynamic Model that Allows for Equity

Main message: debt buyback mechanism **still holds** with external equity financing.



When do firms issue equity? when cash is scarce and when debt is not too high

Empirical Evidence

- Sample: all US corporate bonds from FISD from 2000 to 2024
  - firm-year panel: 31,705 firm-years, 45, 109 bonds
- TCJA as a **liquidity shock**: it released foreign-held cash for domestic use.
  - before TCJA: U.S. multinationals had tax incentives to keep foreign cash abroad;
  - TCAJ: made repatriation much cheaper at the margin
- The dynamic model predicts a state-space payout:
  - $\bar{W}(F) < W < W^{\star}(F) \Rightarrow$ debt buyback
  - $W \geq W^{\star}(F) \Rightarrow$ equity payout
- Sort firms into quartiles by cash shock (average foreign income-to-sales):
  - Q3-windfall moves firms into buyback region
  - Q4-windfall pushes firms past payout boundary

Empirical Evidence

	𝟙(Tender Offers)(%)		Frac(Tender Offers)(%)		Frac(Equity Repurchase)(%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Quartile 3 PIFO × Post	4.714**	4.799**	0.699**	0.709**	0.071	0.047
	(1.990)	(1.990)	(0.324)	(0.324)	(0.303)	(0.301)
Quartile 4 PIFO × Post	3.158*	3.087*	0.279	0.272	0.543**	0.574**
	(1.841)	(1.841)	(0.286)	(0.286)	(0.275)	(0.277)
Leverage(t-1)		6.105**		0.721		-2.301***
		(2.908)		(0.527)		(0.477)
Adjusted R^2	0.103	0.103	0.054	0.054	0.288	0.295
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
NAICS3-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,598	5,571	5,595	5,568	5,468	5,453

Moderate liquidity shocks ⇒ bond buybacks; Large liquidity shocks ⇒ equity payouts.

Outline

Comment 1: Alternative Explanations

Comment 2: What about nonsmooth equilibria?

Comment 1: Motivation and Framing

- This paper is currently framed around the non-buyback puzzle generated in DeMarzo and He (2021).

- Indeed, in a very skeptical world with no commitment, adding liquidity friction can generate buybacks.

Comment 1: Motivation and Framing

- This paper is currently framed around the non-buyback puzzle generated in DeMarzo and He (2021).
- Indeed, in a very skeptical world with no commitment, adding liquidity friction can generate buybacks.
- However, there are already alternative mechanisms to generate buybacks.
- For example, Malenko and Tsoy (2026) show the reputation concern can restore commitment and generate buybacks.
  - Markov perfect equilibrium vs. non-MPE

Comment 1: Motivation and Framing

- This paper is currently framed around the non-buyback puzzle generated in DeMarzo and He (2021).
- Indeed, in a very skeptical world with no commitment, adding liquidity friction can generate buybacks.
- However, there are already alternative mechanisms to generate buybacks.
- For example, Malenko and Tsoy (2026) show the reputation concern can restore commitment and generate buybacks.
  - Markov perfect equilibrium vs. non-MPE
- **Suggestion**: downplay the puzzle and emphasize the importance of a **general framework** featuring both cash liquidity and debt without commitment.

Outline

Comment 1: Alternative Explanations

Comment 2: What about nonsmooth equilibria?

Optimality Condition and Value Function

Solving for a smooth equilibrium is already difficult.

Optimality Condition and Value Function

Solving for a smooth equilibrium is already difficult.

For a given state (F, W), issuing a small amount dF produces two effects:

$$\underbrace{V_F\, dF}_{\text{effect of the additional debt}} + \underbrace{V_W \cdot p\, dF}_{\text{value of the cash raised}} .$$

Optimality Condition and Value Function

Solving for a smooth equilibrium is already difficult.

For a given state (F, W), issuing a small amount dF produces two effects:

$$\underbrace{V_F\, dF}_{\text{effect of the additional debt}} + \underbrace{V_W \cdot p\, dF}_{\text{value of the cash raised}} .$$

For the firm to adjust debt at a finite rate in a smooth equilibrium, these must balance:

$$\boxed{V_F + pV_W = 0.}$$

Idea: given the equilibrium price and continuation value, shareholders have no first-order gain from deviating from the current debt policy.

Optimality Condition and Value Functionk

Value function for the equity holders:

$$V(F_t, W_t) = \max_{\{\Gamma_s, D_s\}_{s\geq t}} \mathbb{E}_t\left[\int_t^{\tau\wedge T} e^{-\rho(s-t)}\, dD_s + e^{-\rho(T-t)}\mathbf{1}_{\{T<\tau\}}\widehat{V}(F_T, W_T)\right]. \tag{12}$$

Rewrite $d\Gamma_s \equiv G_t dt$, under the region where the firm doesn't pay out dividends, the HJB is

$$\rho V(F,W) = \max_G \left\{ GV_F + \left[(1-\pi)(\mu - cF) + pG\right]V_W + \frac{1}{2}(1-\pi)^2\sigma^2 V_{WW} + \lambda\left[\widehat{V}(F,W) - V(F,W)\right]\right\}. \tag{23}$$

Optimality Condition and Value Functionk

Value function for the equity holders:

$$V(F_t, W_t) = \max_{\{\Gamma_s, D_s\}_{s\geq t}} \mathbb{E}_t\left[\int_t^{\tau\wedge T} e^{-\rho(s-t)}\, dD_s + e^{-\rho(T-t)}\mathbf{1}_{\{T<\tau\}}\widehat{V}(F_T, W_T)\right]. \tag{12}$$

Rewrite $d\Gamma_s \equiv G_t dt$, under the region where the firm doesn't pay out dividends, the HJB is

$$\rho V(F,W) = \max_G \left\{ GV_F + \left[(1-\pi)(\mu - cF) + pG\right]V_W + \frac{1}{2}(1-\pi)^2\sigma^2 V_{WW} + \lambda\left[\widehat{V}(F,W) - V(F,W)\right]\right\}. \tag{23}$$

We can characterize the value function, without even knowing debt issuance and buyback:

$$(\rho+\lambda)V = (1-\pi)(\mu - cF)V_W + \frac{1}{2}(1-\pi)^2\sigma^2 V_{WW} + \lambda W. \tag{24}$$

- For a given F, this is an ODE for V w.r.t. W!

$\Rightarrow$ solve for the equity holders' value function $V(F,W)$ ✓

Optimality Condition and Value Functionk

Value function for the equity holders:

$$V(F_t, W_t) = \max_{\{\Gamma_s, D_s\}_{s \geq t}} \mathbb{E}_t \left[\int_t^{\tau \wedge T} e^{-\rho(s-t)} \, dD_s + e^{-\rho(T-t)} \mathbf{1}_{\{T<\tau\}} \widehat{V}(F_T, W_T) \right]. \tag{12}$$

Rewrite $d\Gamma_s \equiv G_t dt$, under the region where the firm doesn't pay out dividends, the HJB is

$$\rho V(F, W) = \max_G \left\{ G V_F + \left[(1-\pi)(\mu - cF) + pG \right] V_W + \frac{1}{2}(1-\pi)^2 \sigma^2 V_{WW} + \lambda \left[\widehat{V}(F, W) - V(F, W) \right] \right\}. \tag{23}$$

We can characterize the value function, without even knowing debt issuance and buyback:

$$(\rho + \lambda) V = (1-\pi)(\mu - cF) V_W + \frac{1}{2}(1-\pi)^2 \sigma^2 V_{WW} + \lambda W. \tag{24}$$

- For a given F, this is an ODE for V w.r.t. W!

$\Rightarrow$ solve for the equity holders' value function $V(F, W)$ ✓

- this technique has also been used in DeMarzo and He (2021)

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \; \overbrace{-\left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \overbrace{- \left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \overbrace{- \left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow
2. Cash-carry wedge [Why equity holders might want to repurchase]

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \overbrace{- \left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow
2. Cash-carry wedge [Why equity holders might want to repurchase]
   - ρ: equity holders are impatient, issuing an extra unit of debt without distributing cash (on the B/S) is a cost

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \overbrace{- \left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow
2. Cash-carry wedge [Why equity holders might want to repurchase]
   - ρ: equity holders are impatient, issuing an extra unit of debt without distributing cash (on the B/S) is a cost
   - $\frac{\lambda(V_W - 1)}{V_W}$: termination reduces the cash value from V_W to 1 with prob. $\lambda \rightarrow$ termination destroys the liquidity premium (expected termination loss)

Optimal Debt Policy

Having characterized the value function $V(F, W)$, the optimal debt policy is obtained from the equilibrium condition:

$$G_t = \frac{\overbrace{\pi c}^{\text{tax benefit}} \overbrace{- \left[\rho + \lambda \frac{V_W - 1}{V_W}\right] p}^{\text{cash-carry wedge}} + \overbrace{(1-\pi)^2 \sigma^2 \cdot p_W \cdot \frac{-V_{WW}}{V_W}}^{\text{precautionary liquidity}}}{-(p_F + p \cdot p_W)} = \frac{\text{net benefits}}{\text{adverse price impact}}. \tag{34}$$

1. Tax benefit: coupon payments are tax deductible $\rightarrow$ debt tax shield benefit flow
2. Cash-carry wedge [Why equity holders might want to repurchase]
   - ρ: equity holders are impatient, issuing an extra unit of debt without distributing cash (on the B/S) is a cost
   - $\frac{\lambda(V_W - 1)}{V_W}$: termination reduces the cash value from V_W to 1 with prob. $\lambda \rightarrow$ termination destroys the liquidity premium (expected termination loss)
3. Precautionary motive: equity value is concave in cash ($V_{WW} < 0$) because cash is more valuable when the firm is closer to running out of cash

Comment 2: Solution Logic and Nonsmooth Eq

This paper uses a **conjecture-and-verify** method to construct a smooth equilibrium.

i. suppose the firm only adjusts debt smoothly, we can construct a smooth equilibrium
ii. in the smooth eq, indeed smooth debt adjustment is optimal (*Lemma 2*)

Comment 2: Solution Logic and Nonsmooth Eq

This paper uses a **conjecture-and-verify** method to construct a smooth equilibrium.

i. suppose the firm only adjusts debt smoothly, we can construct a smooth equilibrium
ii. in the smooth eq, indeed smooth debt adjustment is optimal (*Lemma 2*)

What if the firm can adjust the debt rapidly? The paper doesn't offer a clear overall picture.

Comment 2: Solution Logic and Nonsmooth Eq

This paper uses a **conjecture-and-verify** method to construct a smooth equilibrium.

i. suppose the firm only adjusts debt smoothly, we can construct a smooth equilibrium
ii. in the smooth eq, indeed smooth debt adjustment is optimal (*Lemma 2*)

What if the firm can adjust the debt rapidly? The paper doesn't offer a clear overall picture.

Section 4.2 shows both the **robustness** and **limitation** of smooth eq:

- Robustness: if outside equity is too expensive, the baseline smooth eq remains valid
- Limitation: if equity issuance can be nonsmooth, the smooth debt adjustment eq cannot sustain

Comment 2: Solution Logic and Nonsmooth Eq

This paper uses a **conjecture-and-verify** method to construct a smooth equilibrium.

i. suppose the firm only adjusts debt smoothly, we can construct a smooth equilibrium
ii. in the smooth eq, indeed smooth debt adjustment is optimal (*Lemma 2*)

What if the firm can adjust the debt rapidly? The paper doesn't offer a clear overall picture.

Section 4.2 shows both the **robustness** and **limitation** of smooth eq:

- Robustness: if outside equity is too expensive, the baseline smooth eq remains valid
- Limitation: if equity issuance can be nonsmooth, the smooth debt adjustment eq cannot sustain

$\Rightarrow$ Q: What would nonsmooth equilibrium look like with cash management?

Recap

- A very solid theoretical paper with supporting empirical evidence
- **Punchline:** Debt buybacks are **puzzling** under leverage-ratchet logic $\Rightarrow$ **incorporating the liquidity friction** resolves this puzzle.
- A promising research agenda: future research could further explore the interaction between cash and other frictions, such as transaction costs, reputation concerns, or agency problems.
- I learned a lot from reading this paper!